

Initial Technology Choices

Frontend: Next.js

In a previous full-stack project I did for CS316, the focus was more on how the backend and database connected to each other and how the backend retrieved and stored data in the database. There was less focus on the frontend, so I really haven't had much experience developing an interactive UI. I wanted to learn more about frontend development, and I had read and heard that React and React frameworks can make building interactive user interfaces more efficient by allowing the UI to be broken into reusable components. I chose to use Next.js over React on its own, not for any particular reason. I thought having the ability to explicitly control server-side rendering and client-side rendering was a plus, but ultimately, I don't expect it to be a major factor in the early stages of the project.

Backend: FastAPI (?)

I wanted to use a Python backend because I am familiar with Python, and using the language would build on what I already learned from my previous full-stack project. I am most comfortable with Python, and using a separate Python backend also lets me keep the backend and frontend as separate parts of the application rather than learning Next.js's backend/API functionality at the same time. The two tools I was deciding between were Flask and FastAPI.

Earlier, I planned on using FastAPI because I read that it had built-in utilities for things like request validation and implementing authentication and authorization. The fact that FastAPI is specifically designed around building APIs also caught my attention, especially because the backend will primarily communicate with the Next.js frontend through API requests. There were also many tutorials and examples showing how to connect a React/Next.js frontend to a FastAPI backend, so I knew there would be resources available when it came to integrating the two.

In hindsight, it might have been a good idea to stick with Flask, since I have used Flask in a previous project and am already familiar with it. Flask is also a more mature framework, so there is an argument for using something I already know rather than introducing another framework. I'm not sure if I'll want to switch back to Flask later down the line, but right now I am planning to use FastAPI.

Database Management System: PostgreSQL

I chose PostgreSQL because that is what I used for my previous full-stack application, so I am already comfortable working with it. Using PostgreSQL also allows me to build on what I learned from that project rather than having to learn a new database system